A CASE-STUDY BASED REPORT

ON

Multi-Class Customer Category Prediction Using Multinomial Logistic Regression for E-Commerce Customer Segmentation

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Abstract

Customer segmentation is an essential task in e-commerce because it allows a business to understand how different customers behave and to serve each of them with the right offers, services and communication. This case study, "Multi-Class Customer Category Prediction Using Multinomial Logistic Regression for E-Commerce Customer Segmentation", develops a supervised machine-learning model that assigns every customer to one of four categories — Budget, Regular, Premium and VIP — using demographic information, purchasing behaviour and transaction history.

The dataset of 2,000 customers and 9 predictor attributes (age, gender, annual income, purchase frequency, spending score, average order value, tenure, discount-usage rate and preferred product category) was cleaned by removing 15 duplicate records and imputing missing values, after which numerical features were standardised and categorical features were one-hot encoded. A multinomial (softmax) logistic regression model was trained on 80% of the data and evaluated on the remaining 20%. On the unseen test set the model achieved an accuracy of 94.50%, a macro-averaged F1-score of 0.953, and a five-fold cross-validated accuracy of 93.85%. The model is interpretable through its coefficients and returns class probabilities, which makes it suitable for targeted marketing, personalised recommendations and resource allocation.

Keywords: Customer Segmentation, Multi-Class Classification, Multinomial Logistic Regression, Softmax, E-Commerce, Exploratory Data Analysis.

1. Introduction

1.1 Background

In a competitive e-commerce environment, customers differ widely in their purchasing habits, spending power, product preferences and loyalty. Treating all customers identically wastes marketing budget and weakens the customer experience. Organisations therefore group customers into meaningful categories so that promotions, pricing, service levels and retention strategies can be tailored to each group.

Traditional customer classification relies on manual analysis or on binary classifiers (for example "will buy" versus "will not buy"), which cannot represent more than two customer tiers. Multinomial Logistic Regression is a supervised learning algorithm that generalises logistic regression to more than two classes. It estimates, for every customer, the probability of belonging

to each predefined category and selects the most probable one, which makes it a natural fit for multi-class customer category prediction.

1.2 Motivation for the Project

- Growing customer bases make manual tiering slow, inconsistent and difficult to scale.
- Targeted campaigns for the right tier improve conversion and retention while reducing wasted discounts.
- A probabilistic and interpretable model lets managers understand why a customer is placed in a category, which black-box models do not offer.
- Multinomial logistic regression is computationally light, fast to retrain and a strong baseline for tabular business data, which suits a course-level case study.

1.3 Overview of the Work Carried Out

The work followed the standard machine-learning workflow. First, the customer dataset was loaded and inspected for size, data types, missing values and duplicates. Second, exploratory data analysis (EDA) was carried out using class-distribution plots, histograms, box plots, group-wise means and a correlation heatmap. Third, the data was pre-processed by removing duplicates, imputing missing values, encoding categorical variables and standardising numerical variables. Fourth, a multinomial logistic regression model was trained inside a scikit-learn pipeline using a stratified 80:20 train-test split. Finally, the model was evaluated using accuracy, precision, recall, F1-score, a confusion matrix and five-fold cross-validation, and its coefficients were interpreted.

1.4 Scope and Responsible-Use Statement

Scope. The study covers the prediction of one of four customer categories from tabular customer attributes using a single, interpretable algorithm. It does not cover real-time deployment, recommender systems, or comparison with every possible classifier.

Responsible use. Customer data is personal data. In a real deployment the data must be collected with consent, stored securely and anonymised (the Customer ID is dropped before modelling). Attributes such as gender and age must not be used to deny services or offer unfair prices; they are retained here only as descriptive features and the model should be audited for bias across groups before use. Predicted categories are intended to support marketing decisions, and a human should review any decision that materially affects a customer.

2. Problem Statement

2.1 Case-Study Scenario

An online retail company serves thousands of customers through its website and mobile application. The marketing team currently assigns customers to loyalty tiers using fixed spending thresholds that are reviewed only once a year. As a result, fast-growing customers stay in a low tier for too long, while some lapsed customers continue to receive premium offers. The company holds records of each customer's demographics, income, order frequency, order value, tenure and discount usage, and wants an automated, data-driven way to place every customer in the correct category: Budget, Regular, Premium or VIP.

2.2 Central Case-Study Question

Can a multinomial logistic regression model, trained on demographic and purchasing-behaviour attributes, accurately and interpretably predict which of four categories (Budget, Regular, Premium, VIP) a customer belongs to, and which attributes drive that prediction?

2.3 Formal Problem Definition

Let the dataset be $D = \{(x_i, y_i)\}$, $i = 1 \ldots n$, where $x_i \in \mathbb{R}^d$ is the feature vector of customer i and $y_i \in \{1, 2, 3, 4\}$ is the customer category (1 = Budget, 2 = Regular, 3 = Premium, 4 = VIP). The goal is to learn a function $f: \mathbb{R}^d \rightarrow \{1,\ldots,4\}$ that minimises the expected misclassification on unseen customers. Under the multinomial logistic model, the probability that customer x belongs to class k is given by the softmax function:

$$P(y = k \mid x) = \exp(w_k \cdot x + b_k) / \Sigma_j \exp(w_j \cdot x + b_j), \quad k = 1,\ldots,4$$

The parameters (w_k, b_k) are estimated by minimising the L2-regularised cross-entropy (negative log-likelihood) loss, and the predicted class is $\hat{y} = \text{argmax}_k P(y = k \mid x)$.

2.4 Objectives that Guide the Study

1. To collect and understand a customer dataset containing demographic, behavioural and transactional attributes.
2. To clean and pre-process the data (duplicates, missing values, encoding and scaling) so that it is suitable for modelling.
3. To perform exploratory data analysis and identify the attributes that separate the four customer categories.
4. To build a multinomial logistic regression classifier for four customer categories.

5. To evaluate the model using accuracy, precision, recall, F1-score, confusion matrix and cross-validation.
6. To interpret the model coefficients and recommend how the predictions can support marketing decisions.

2.5 Assumptions and Limitations

Assumptions:

- The category labels (Budget, Regular, Premium, VIP) are correct and were assigned by the business.
- The relationship between the standardised features and the log-odds of each class is approximately linear.
- Records are independent of one another, and the training data is representative of future customers.
- Missing values occur at random and can be reasonably replaced by the median.

Limitations:

- A linear decision boundary cannot capture complex feature interactions; tree-based or neural models may fit non-linear patterns better.
- Features are on different scales and some (such as order value) are right-skewed; only standardisation was applied, without further transformation.
- The dataset used here is a simulated dataset of 2,000 customers prepared for this case study (see Section 3.1); results on real company data may differ and should be re-validated.
- The model uses a snapshot of each customer and does not model how customers move between categories over time.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The study uses a customer category prediction dataset in CSV format. It contains 2,015 raw records and 11 columns. The dataset used in this report is a simulated e-commerce dataset, generated with a fixed random seed (42) so that the experiment is fully reproducible; it was designed to mimic realistic customer behaviour, including missing values and a few duplicate rows. The same code runs unchanged on a real company CSV that has the same columns. The attributes are described in Table 1.

Attribute	Type	Description
Customer_ID	Identifier	Unique customer code (dropped before modelling)
Age	Numerical	Age of the customer in years
Gender	Categorical	Male / Female
Annual_Income_k	Numerical	Annual income in thousands
Purchase_Frequency	Numerical	Number of purchases in the last 12 months
Spending_Score	Numerical	Score from 1 to 100 based on spending behaviour
Avg_Order_Value	Numerical	Average value of one order
Tenure_Months	Numerical	Months since the customer registered
Discount_Usage_Rate	Numerical	Share of orders placed using a discount (0 to 1)
Preferred_Category	Categorical	Electronics, Fashion, Home & Kitchen, Beauty, Grocery
Customer_Category	Target (4 classes)	Budget, Regular, Premium, VIP

Table 1: Description of dataset attributes

3.2 Purpose of EDA

Exploratory data analysis is carried out before modelling to understand the structure and quality of the data. In this study EDA is used to (i) detect missing values and duplicates, (ii) check whether the four classes are balanced, (iii) understand the distribution and spread of each numerical feature, (iv) find which features differ most between categories, and (v) detect strongly correlated features that might affect a linear model.

3.3 Preparation Steps Actually Performed

1. **Loading and inspection:** the CSV was loaded with Pandas; its shape was 2,015 rows × 11 columns.
2. **Duplicate removal:** 15 duplicate records (identical in every attribute except the ID) were removed, leaving 2,000 customers.
3. **Missing values:** Age (53), Annual_Income_k (88) and Spending_Score (54) contained missing entries. They were filled with the median of the training data, which is robust to outliers.
4. **Dropping the identifier:** Customer_ID was removed because it carries no predictive information.
5. **Encoding:** Gender and Preferred_Category were converted to numerical form using one-hot encoding with the first level dropped to avoid redundant columns.
6. **Scaling:** the seven numerical features were standardised (zero mean, unit variance) with StandardScaler, which helps the solver converge and makes coefficients comparable.
7. **Train-test split:** the data was split 80:20 with stratification on the target, giving 1,600 training and 400 test records. All imputers and scalers were fitted on the training data only, to avoid data leakage.

3.4 EDA Exploration – Implementation

```
import pandas as pd, numpy as np
import matplotlib.pyplot as plt, seaborn as sns

df = pd.read_csv('customer_data.csv')
print(df.shape); df.info()
print(df.isna().sum())               # missing values
print(df.duplicated().sum())         # duplicates
print(df['Customer_Category'].value_counts())
print(df.describe())
sns.countplot(x='Customer_Category', data=df)
df[num_cols].hist(bins=30)
sns.boxplot(x='Customer_Category', y='Annual_Income_k', data=df)
sns.heatmap(df[num_cols].corr(), annot=True, cmap='coolwarm')
```

Class distribution. Figure 1 shows the number of customers in each category. Regular is the largest class and VIP the smallest, so the data is moderately imbalanced; stratified splitting keeps these proportions in both the training and the test sets.

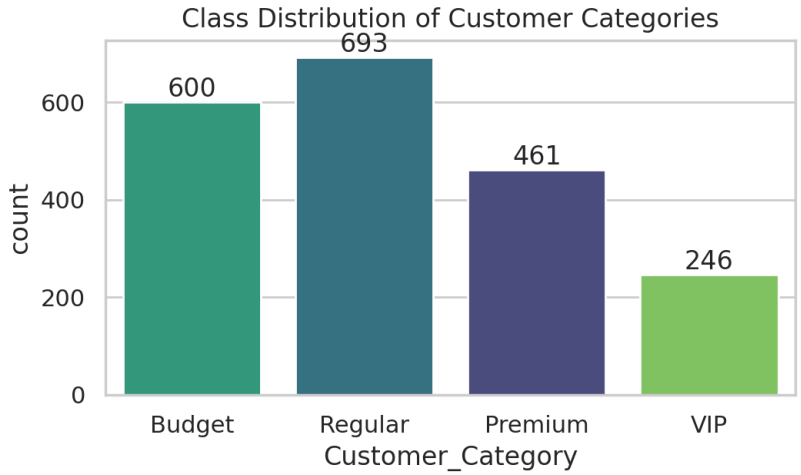


Figure 1: Class distribution (Budget 600, Regular 693, Premium 461, VIP 246)

Feature distributions. The histograms in Figure 2 show that Age, Spending_Score and Tenure are fairly spread out, whereas Annual_Income_k and Avg_Order_Value are right-skewed because a small number of high-value customers spend much more than the rest.

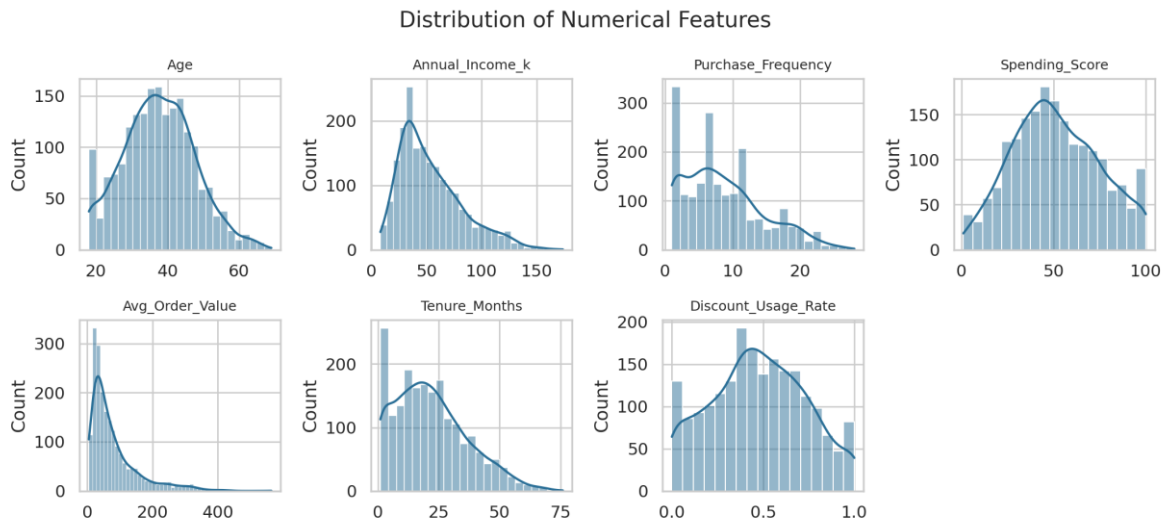


Figure 2: Distribution of numerical features

Descriptive statistics. Table 2 summarises the central tendency and spread of the numerical features after removing duplicates.

Feature	Count	Mean	Std	Min	Median	Max
Age	1948	37.62	9.95	18.0	38.0	69.0
Annual_Income_k	1912	53.94	28.91	8.0	47.3	173.9
Purchase_Frequency	2000	8.68	5.92	1.0	8.0	28.0
Spending_Score	1946	51.31	23.57	1.0	49.0	100.0
Avg_Order_Value	2000	80.45	74.87	5.0	54.24	564.17
Tenure_Months	2000	21.82	14.87	1.0	20.0	76.0
Discount_Usage_Rate	2000	0.47	0.26	0.0	0.47	1.0

Table 2: Descriptive statistics of numerical features

Differences between categories. The box plots in Figure 3 and the group means in Table 3 show a clear, ordered separation of the four categories on income, spending score and purchase frequency: values rise steadily from Budget to VIP, with only slight overlap between neighbouring classes (mainly Budget–Regular and Regular–Premium).

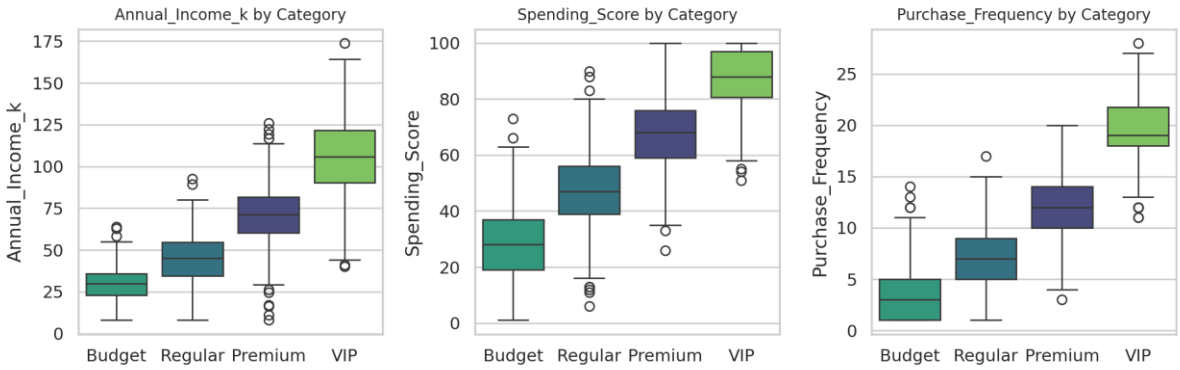


Figure 3: Income, spending score and purchase frequency by customer category

Feature	Budget	Regular	Premium	VIP
Age	35.04	37.57	38.9	41.66
Annual_Income_k	29.85	44.88	71.19	104.91
Purchase_Frequency	3.44	7.13	12.06	19.53
Spending_Score	28.21	47.67	67.75	86.79
Avg_Order_Value	25.01	53.86	110.14	234.91
Tenure_Months	9.06	18.41	29.69	47.78
Discount_Usage_Rate	0.7	0.5	0.3	0.19

Table 3: Mean value of each feature by customer category

Correlation analysis. The heatmap in Figure 4 shows that income, purchase frequency, spending score, average order value and tenure are strongly and positively correlated with one another, while discount usage is negatively correlated with them. Age is only weakly related to the other features.

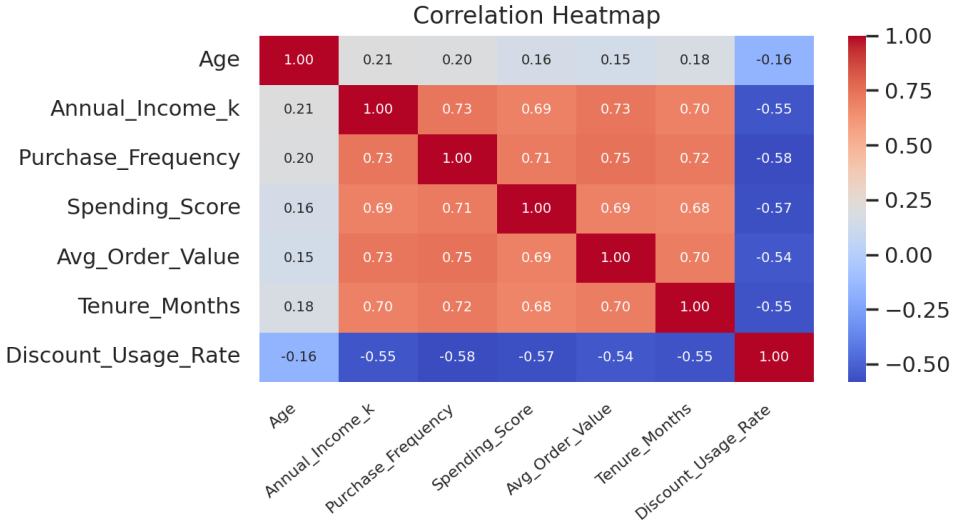


Figure 4: Correlation heatmap of numerical features

3.5 Summary of EDA Findings

- The dataset is of good quality: only 15 duplicates and missing values in three columns (Age 53, Income 88, Spending Score 54), all of which were handled during preparation.
- The four classes are moderately imbalanced (VIP is about 12% of customers), so accuracy alone is not sufficient; precision, recall and F1-score per class are also reported.
- Purchase frequency (r = 0.85), income (r = 0.83), spending score (r = 0.83) and average order value (r = 0.82) have the strongest association with the ordered category label, while age is weak (r = 0.21).
- Discount usage falls as the category rises (Budget 0.70 to VIP 0.19), meaning that lower tiers are more discount-driven.

- Many predictors are correlated with each other (multicollinearity), so L2 regularisation is appropriate to keep coefficient estimates stable.
- Neighbouring categories overlap slightly, so the few expected errors should occur between adjacent tiers and not between distant ones.

4. Algorithm Selection and Implementation

4.1 Why Multinomial Logistic Regression

The target has four categories, so a multi-class classifier is required. Multinomial logistic regression was chosen because (a) it models all classes jointly through the softmax function and produces calibrated class probabilities, (b) its coefficients can be read directly to explain which attributes push a customer towards a category, (c) it trains quickly and scales to large customer bases, and (d) it is a strong, standard baseline for tabular data with mostly linear class boundaries, as seen in the EDA.

4.2 Working of the Algorithm

1. For each class k, compute a linear score $z_k = w_k \cdot x + b_k$.
2. Convert the four scores to probabilities with the softmax function: $p_k = \exp(z_k) / \Sigma_j \exp(z_j)$. The probabilities are positive and sum to 1.
3. Measure the error with the cross-entropy loss, $L = -(1/n) \Sigma_i \log p_{i,y_i}$, plus an L2 penalty ($\lambda\|W\|^2$) to avoid over-fitting.
4. Minimise the loss with the L-BFGS optimiser to obtain the weights.
5. For a new customer, predict the class with the highest probability.

4.3 Software Requirements

- **Operating system:** Windows 10/11 or Linux
- **Language:** Python 3.x
- **Environment:** Jupyter Notebook or Visual Studio Code
- **Libraries:** NumPy (numerical computation), Pandas (data handling), Matplotlib and Seaborn (visualisation), Scikit-learn (pre-processing, model, metrics) and Joblib (saving the trained model)
- **Dataset:** Customer Category Prediction dataset (CSV)

4.4 Implementation

```
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

```
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix
import joblib

X = df.drop(columns=['Customer_ID', 'Customer_Category'])
y = df['Customer_Category']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42)

pre = ColumnTransformer([
  ('num', Pipeline([('imp', SimpleImputer(strategy='median')),
                    ('sc', StandardScaler())]), num_cols),
  ('cat', OneHotEncoder(drop='first'), ['Gender', 'Preferred_Category'])])

model = Pipeline([('prep', pre),
                  ('clf', LogisticRegression(solver='lbfgs', max_iter=1000))])
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(cross_val_score(model, X, y, cv=5))
joblib.dump(model, 'customer_model.joblib')
```

In scikit-learn, LogisticRegression with the lbfgs solver automatically fits a multinomial (softmax) model when the target has more than two classes. Wrapping the pre-processing and the classifier in one Pipeline ensures that the imputation and scaling learnt from the training data are applied identically to the test data and to any new customer.

4.5 Results and Evaluation

The model reached an accuracy of **94.50%** on the 400 unseen test customers (training accuracy 94.50%). Because training and test accuracy are almost identical, the model is not over-fitting. Table 4 gives the per-class precision, recall and F1-score.

Class	Precision	Recall	F1-score	Support
Budget	0.949	0.933	0.941	120
Regular	0.915	0.935	0.925	139
Premium	0.957	0.957	0.957	92
VIP	1.000	0.980	0.990	49
Macro average	0.955	0.951	0.953	400
Weighted average	0.945	0.945	0.945	400

Table 4: Classification report on the test set

The confusion matrix in Figure 5 shows where the errors occur. Of 400 test customers, only 22 were misclassified, and every error is between adjacent categories (for example Budget↔Regular or Regular↔Premium); no Budget customer was predicted as Premium or VIP. VIP customers are identified almost perfectly (48 of 49 correct), which is valuable because they are the most profitable group.

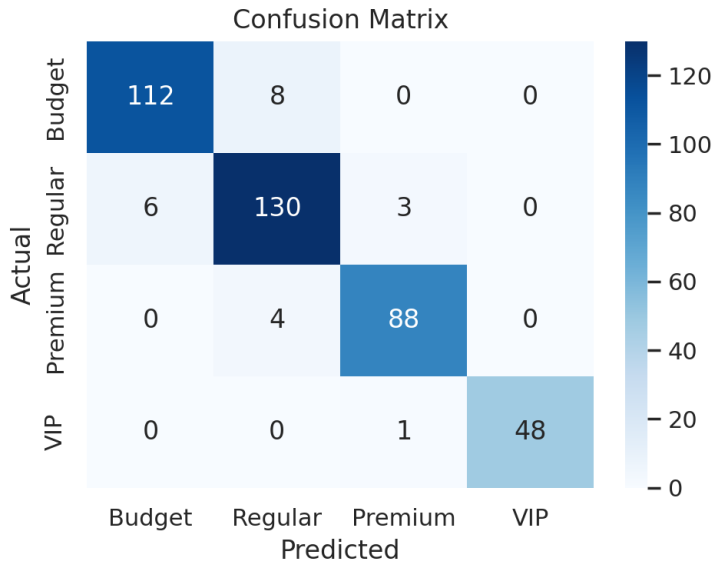


Figure 5: Confusion matrix on the test set

Cross-validation. Five-fold cross-validation gave accuracies of 94.25%, 95.25%, 92.50%, 93.75%, 93.50%, with a mean of 93.85%, close to the test accuracy, which indicates that the result is stable and not dependent on one particular split.

4.6 Interpretation of Coefficients

Because the features were standardised, the coefficients in Figure 6 can be compared directly: a positive value means that a higher value of the feature increases the probability of that class. Average order value, purchase frequency, income, spending score and tenure have large positive coefficients for VIP and Premium and large negative coefficients for Budget, whereas discount-usage rate shows the opposite sign. Age and the one-hot encoded gender and product-preference columns contribute very little, which agrees with the EDA and suggests that behaviour, not demographics, drives category membership.

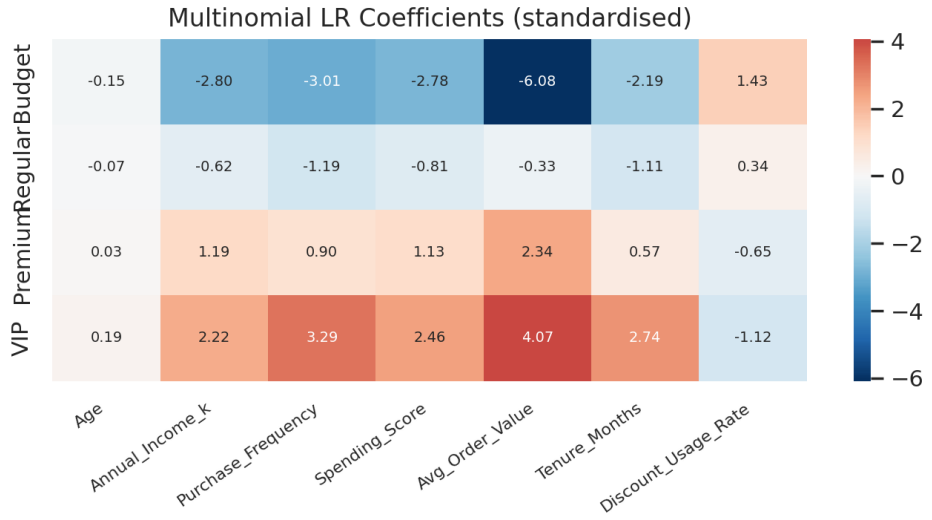


Figure 6: Coefficients of the multinomial logistic regression model

4.7 Business Use of the Model

- **VIP / Premium:** offer early access, loyalty rewards and dedicated support to retain the most valuable customers.
- **Regular:** send cross-sell recommendations and targeted offers to move customers towards the Premium tier.
- **Budget:** use limited-period discounts and low-cost engagement to increase purchase frequency.
- **Probabilities:** customers whose top two class probabilities are close can be flagged as near the boundary of a tier and be targeted for upgrade campaigns.

5. Conclusion

This case study applied multinomial logistic regression to predict the category — Budget, Regular, Premium or VIP — of e-commerce customers from demographic and purchasing-behaviour attributes. After cleaning, imputation, encoding and scaling, the model achieved 94.50% accuracy and a macro F1-score of 0.953 on unseen data, with a cross-validated accuracy of 93.85%. All misclassifications occurred between neighbouring categories, and VIP customers were identified with 100% precision and about 98% recall.

The EDA and the coefficients agree that purchase frequency, average order value, income, spending score and tenure are the strongest drivers of customer category, and that discount dependence is typical of lower tiers. The model is simple, fast, interpretable and probabilistic, so it can be used directly by marketing teams for targeted campaigns, personalised recommendations and better allocation of resources.

Future work: (i) validate the model on real transactional data, (ii) compare it with Random Forest, Gradient Boosting and Support Vector Machines, (iii) add recency-frequency-monetary (RFM) features and time-based behaviour, (iv) tune the regularisation strength with grid search, and (v) audit the model for fairness across gender and age groups before deployment.

Note: the dataset in this report is simulated; the numerical results describe that dataset and should be regenerated if a real dataset is used.

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Signature of the Team members